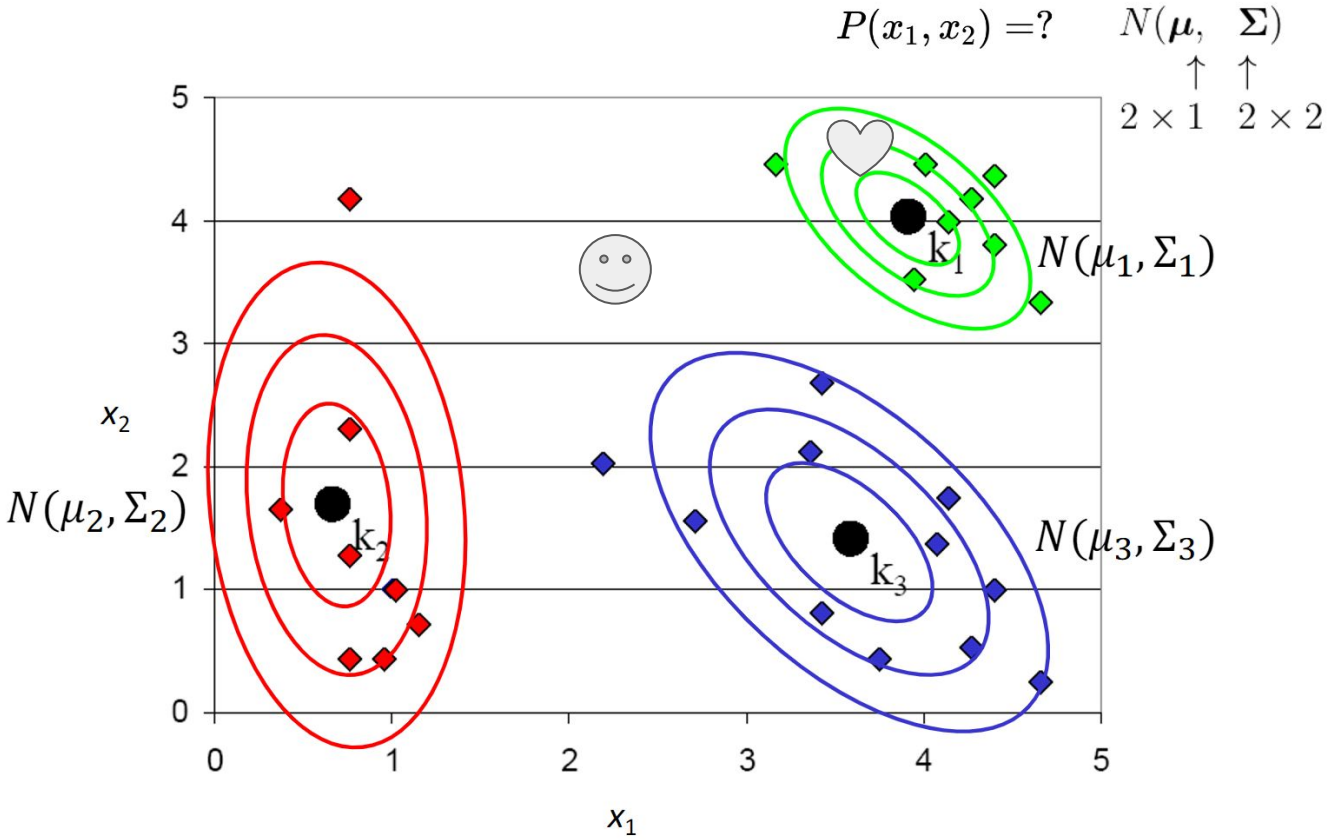
Recitation 3/21

EM Algorithm Implementation

Soft k-means clustering



EM Algorithm Overview

- Expectation step (E step) (assign clusters)
  - Compute expected likelihood wrt (latent cluster labels)

$$E_{p(c^n|x^n,\theta^{(t)})}\left[\log \prod_{n=1}^{N} p(x^n, c^n)\right]$$

  - $\theta^{(t)} = \{\pi_i, \mu_i, \Sigma_i\}_{i=1}^{K}$ are current estimates of parameters

- Maximization step (M step) (estimate Gaussian mixture parameters)
  - Find the parameters that maximize this quantity

$$\theta^{(t+1)} = \text{argmax}_\theta E_{p(c^n|x^n,\theta^{(t)})}\left[\log \prod_{n=1}^{N} p(x^n, c^n)\right]$$

- EM algorithm is guaranteed to converge: the log likelihood of the observed data is guaranteed to increase in each iteration

EM Questions

Why do we have iterations? Why doesn’t the argmax part just solve it?

Does the initialization matter?

Supervised or unsupervised?

EM Questions

Why do we have iterations? Why doesn't the argmax part just solve it?

Because the argmax depends on the expectation, which depends on the argmax.

Does the initialization matter?

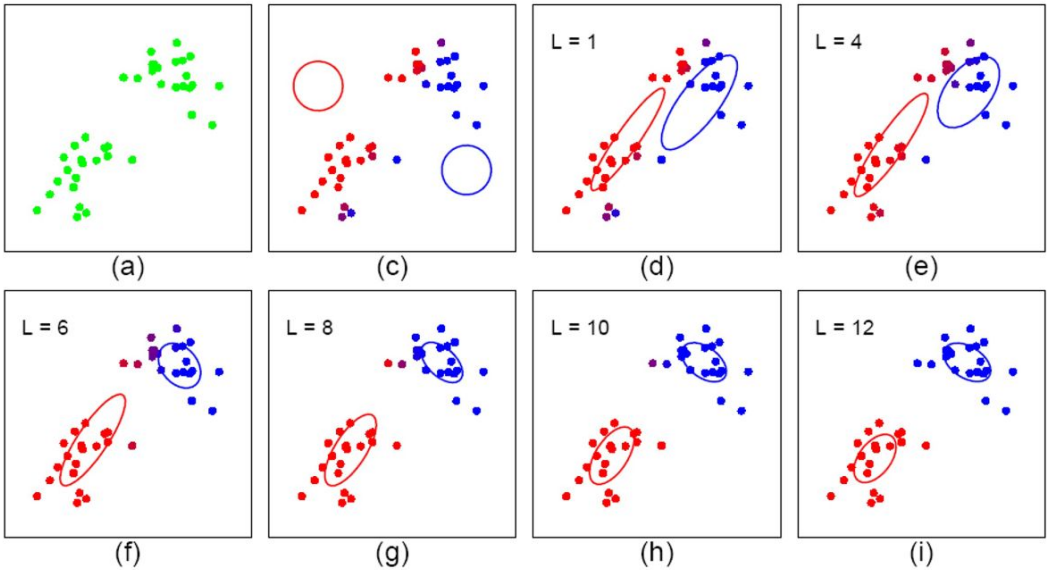
Yes. Converges to local minimum.

Supervised or unsupervised?

Unsupervised

EM in the context of soft k-means clustering

- Start:
  - "Guess" the centroid μ_i and covariance Σ_i of each of the K clusters
- Loop



EM for GMM Visualization

https://colab.research.google.com/drive/1-g5Hoo3rIIL26nPSjd1Htntfcp3-zZPs#scrollTo=IlRzyES2Jlsp

Multivariate Normal PDF

Input: Data point, Mean Vector, Covariance Matrix

Output: Probability of Data given Mean Vector and Covariance Matrix

```
Input:
    - x: (d,) numpy array
    - mean: (d,) numpy array; the mean vector
    - cov: (d,d) numpy array; the covariance matrix
Output:
    prob - a scaler
```

$$f(\mathbf{x};\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^n \det(\boldsymbol{\Sigma})}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

Hidden Matrix

Hidden matrix (n, k) contains the probabilities for each cluster (k), that each data point (n) belongs to that cluster.

Pseudocode:

Compute cluster label probability $P(c^n = i \mid x^n)$ for each data point given the current parameters $\{\pi_i, \mu_i, \Sigma_i\}_{i=1}^K$

$$P(c^n = i \mid x^n) = \frac{p(x^n \mid c^n = i)\, p(c^n = i)}{\sum_{i=1}^K p(x^n \mid c = i) p(c^n = i)}$$

1. For each data point:
    a. For each cluster
        i. Compute the likelihood that that point belongs to that cluster. **How?**
        ii. Multiply that by the mixture proportion for the cluster.
        iii. Store it in the hidden matrix.
2. Normalize so that probabilities sum to 1

UpdateMixProps

Mixing proportions (k) is the default probability for each point to be assigned to each cluster.

Pseudocode:

1. Given a hidden matrix that is normalized, the mixing_props is just the mean across rows.
    a. ie, a k-dimensional array containing the sum of probabilities that each point belongs in that cluster. Like [0.3, 0.3, 0.4]

Compute parameters given cluster labels $P(c^n = i \mid x^n)$

– Mixing proportions

$$\pi_i = p(c = i) = \frac{\sum_{n=1}^{N} p(c^n = i \mid x^n)}{N}$$

UpdateMeans

Updates the means (k, d) which stores the new mean vectors.

Pseudocode:

1. For each cluster:
    a. Compute the weighted sum of the data points in each mixture component
        i. The weights are the responsibilities to the ith component
    b. Compute the sum of the weighted data points.
    c. Normalize by sum of probabilities for that cluster

$$\mu_i = \frac{\sum_{n=1}^{N} x^n p(c^n = i \mid x^n)}{\sum_{n=1}^{N} p(c^n = i \mid x^n)}$$

UpdateCovars

Update the covariances (d,d) for each cluster

Pseudocode:

1. For each cluster:
    a. For each data point:
        i. Compute the deviation of the datapoint from the mean of the cluster
    b. Using each deviation vector, compute the covariance matrix
    c. Weigh this covariance matrix by the responsibility of each data point to that cluster
    d. Normalize by sum of probabilities

$$\Sigma_i = \frac{\sum_{n=1}^{N}(x^n-\mu_i)(x^n-\mu_i)^T p(c^n=i\,|x^n)}{\sum_{n=1}^{N} p(c^n=i\,|x^n)}$$

GMM

Run the whole algorithm

Pseudocode:

1. For each iteration:
    a. Calculate the hidden matrix
    b. Update the mixing proportions
    c. Update the means
    d. Update the covariances
    e. If the current likelihood is similar enough to the previous likelihood, i.e. converged.
        i. Break
        ii. Clusters are what they are currently set to.

Initializations

What initializations do we start with?

Initializations

What initializations do we start with?

- **Mixing proportions [0.3, 0.3, 0.4]**
- **Means [whatever]**
- **Covariances [whatever]**

Log Likelihood Plot

What should it look like?

Log Likelihood Plot

What should it look like?

Line starting from very negative number, going to less negative number.